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Effect of cordycepin on interleukin-10 production of human peripheral blood mononuclear cells

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Abstract

Therapeutic options for controlling autoimmune diseases are still very limited. Interleukin-10 has been reported to be a promising approach to therapeutic intervention. In the search for a drug which results in the selective upregulation of interleukin-10, we investigated the immunoregulative effects of cordycepin. We have measured interleukin-10 and interleukin-2 secretion of human peripheral blood mononuclear cells that were incubated with cordycepin and assessed the influence of cordycepin on the expression of interleukin-10 mRNA, the proliferative response and the expression of surface markers on T lymphocytes. In addition, the subsets of interleukin-10-secreting cells, the influence of anti-interleukin-10 neutralizing antibody and cytotoxicity of cordycepin were evaluated. Our results suggest that cordycepin has a significantly upregulative effect on interleukin-10 production and interleukin-10 mRNA expression. Interleukin-10-producing cells included CD4+, CD8+, CD19+, CD56+ and CD14+ cells. At the same time, cordycepin inhibited phytohaemagglutinin-induced interleukin-2 production and proliferation of peripheral blood mononuclear cells. A restricted T lymphocyte activation was also reflected by a reduced expression of the surface markers CD25, CD45RO, CD54, CD71 and HLA DR. Anti-interleukin-10 neutralizing antibody could not completely block the suppressive effect of cordycepin on production of interleukin-2. Cordycepin in the effective concentration presented slight cytotoxicity but did not increase apoptosis. These results indicate that cordycepin exerts immunoregulative effects. Further research on it may provide an approach for the development of novel immunomodulatory drugs which directly alter the secretion of cytokines.
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Keywords: Cordycepin; Interleukin-10; Interleukin-2; T lymphocyte activation

1. Introduction

Interleukin-10 plays a pivotal role in the cytokine network. A large body of evidence supports the view that interleukin-10 shows a strong regulative effect on Th1 lymphocytes and antigen-presenting cells. The downregulative effect of interleukin-10 on the expression of MHC II antigens, costimulatory molecules and adhesion molecules on antigen-presenting cells contributes to the inhibition of antigen presentation, resulting in the abrogation of proliferative responses and cytokine production of responding T cells or T cell clones. Interleukin-10 also directly inhibits interleukin-2 and interferon-γ production of T cells, independent of its action on antigen-presenting cells (Fiorentino et al., 1989; Moore et al., 1990, 1993; Mottonen et al., 1998; Bogdan et al., 1991; Willems et al., 1994; Schandene et al., 1994). Moreover, interleukin-10 inhibits the secretion of proinflammatory cytokines, e.g. interleukin-1, interleukin-6, interleukin-8 and tumor necrosis factor-α, as well as reduces the production of inflammatory mediators like nitric oxide, free radicals and prostaglandin (Moore et al., 1993; Armstrong et al., 1996; Cunha et al., 1992; Mertz et al., 1994; Dokka et al., 2001). In the process of doing so, interleukin-10 protects the host against excessive immunopathological and inflammatory processes. Deficiency of interleukin-10 is associated with some diseases such as diabetes mellitus, uveitis and inflammatory bowel disease (Gunnett et al., 2002; Rizzo et al., 1998; Gasche et al., 2000).

Although the role of interleukin-10 in health and disease still requires further understanding, experimental researches (Quinn et al., 2000; Bean et al., 1993; Eppinger et al., 1996; Toto et al., 2000; Nakase et al., 2002) and preclinical tests

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(Shirachi, 1998; Van Deventer et al., 1997; McHutchison et al., 1999) have demonstrated that the administration of exogenous interleukin-10 can ameliorate some autoimmune and inflammatory diseases. The development of agents that can selectively affect specific biological action of interleukin-10 may provide more therapeutic options for treating these diseases (Ross et al., 1997; Bourrie et al., 1996; Hill et al., 1998). Nevertheless, up to now, only few drugs are available in clinical practice to stimulate the secretion of endogenous interleukin-10.

In this study, we investigated the immunoregulative effect of cordycepin, a nucleoside derivative found in fungi of the genus cordyceps (Melling et al., 1972; Hanessian et al., 1966; Kaczka et al., 1964). We succeeded to show that cordycepin clearly increased the expression of interleukin-10 mRNA and the secretion of interleukin-10 protein by human peripheral blood mononuclear cells, and simultaneously interfered with interleukin-2 secretion and the proliferative response. In addition, a distinctly altered set of T cell surface markers underlines the vigorous effect of cordycepin on T lymphocyte activation. Therefore, we hypothesize on the potential of cordycepin as a therapeutic immunoregulatory agent.

2. Material and methods

2.1. Preparation of peripheral blood mononuclear cells and cell culture

Blood samples of 30 ml from peripheral veins of 11 healthy adult volunteers (25–47 years of age) were drawn and heparinized. Peripheral blood mononuclear cells were separated by gradient centrifugation with Ficoll (Biochrom, Berlin, Germany). The peripheral blood mononuclear cells were washed two times with RPMI 1640 (Biochrom) and resuspended at a density of 1.0×10^6 cells/ml in culture medium which consisted of RPMI 1640 supplemented with 10% heat-inactivated human AB serum (PAN Biotech, Aidenbach, Germany), 1 mM Na-pyruvate (Biochrom), 0.05 mM 2-mercaptoethanol (Gibco BRL, Karlsruhe, Germany), 1 × minimum essential medium (MEM) nonessential amino acids (Biochrom), 2 mM L-glutamine (Gibco BRL), 100 IU/ml penicillin and 100 μg/ml streptomycin (Biochrom). Aliquots of the resulting cell suspension were supplemented with a blank and different concentration of cordycepin, 3′-deoxyadenosine (Sigma-Aldrich Chemie, Steinheim, Germany), predissolved in phosphate-buffered saline (PBS). From each suspension of peripheral blood mononuclear cells, 100 μl aliquots were removed into two wells of a 96-well microtiter plates and incubated in 5% CO_2 at 37 °C for 2 h. Subsequently, one aliquot was mixed with 100 μl culture medium, and another one aliquot was mixed with 100 μl culture medium containing 1 μg phytohaemagglutinin. All samples were incubated for a total time period of 24 or 48 h.

2.2. Measurement of cytokine expression

After 24 h of incubation with PBS or phytohaemagglutinin or 24 μg/ml cordycepin or phytohaemagglutinin and 24 μg/ml cordycepin, cells were harvested and total RNA was extracted from the cells using a High Pure RNA Isolation Kit (Roche, Mannheim, Germany). The total RNA was reverse transcribed into cDNA in a 20-μl reaction mixture containing 9 μl total cellular RNA, 1 μl (0.5 μg/μl) (dT)15, 1 μl (40 U/μl) RNAsin, 1 μl (200 U/μl) MoMuLv, 2 μl 10 mM dNTP, 4 μl 25 mM $MgCl_2$ and 2 μl 10 × polymerase chain reaction (PCR) buffer. The cDNA was stored at −20 °C until use. PCR analysis of interleukin-10 mRNA and actin mRNA was performed with a LightCycler instrument (Roche) in a 20-μl reaction volume, containing 1.2 μl 25 mM $MgCl_2$, 2 μl SYBR Green (Roche), 2 μl cDNA, 13.8 μl aqua, 0.5 μl 10 mM 3′-primer and 0.5 μl 10 mM 5′-primer (interleukin-10: 5′-GGA GGA CTT TAA

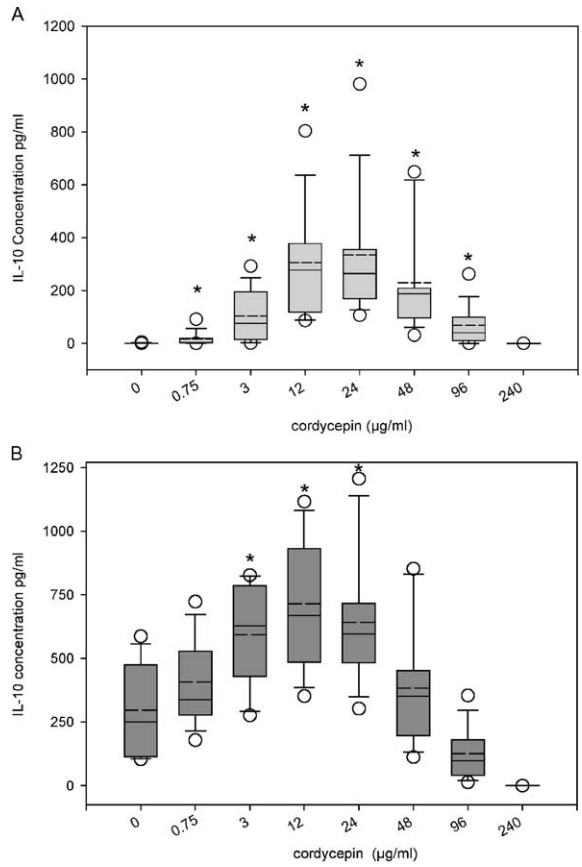

Fig. 1. (A) Without phytohaemagglutinin stimulation, interleukin-10 production of peripheral blood mononuclear cells increased in a dose-dependent manner with increasing cordycepin concentration. Interleukin-10 concentrations in the samples treated with 0.75, 3, 12, 24, 48 and 96 μg/ml of cordycepin were significantly different from that in PBS control ($n = 11$, $*P < 0.05$). (B) Cordycepin enhances the phytohaemagglutinin-induced interleukin-10 production. Cordycepin (12 μg/ml) raised the interleukin-10 product level up to 714.8 ± 78.7 pg/ml versus 295.5 ± 57.7 pg/ml in control. Cordycepin (3, 12 and 24 μg/ml) increased phytohaemagglutinin-induced interleukin-10 production significantly ($n = 11$, $*P < 0.05$).

GGG TTA CCT G-3′, 5′-TGT CAA ACT CAC TCA TGG CTT T-3′; or actin: 5′-TCC TGT GGC ATC CAC GAA ACT-3′, 5′-GAA GCA TTT GCG GTG GAC GAT-3′ as an internal control for normalization). The purity and identity of the PCR products were verified by recording a melting curve of the reaction products and by agarose gel electrophoresis.

Secreted cytokines were measured in supernatants from peripheral blood mononuclear cell cultures after 48 h. Supernatants were harvested by centrifugating cell cultures 15 min, $1500 \times g$ at 4 °C, and were stored at −80 °C until measurement. Secreted interleukin-10 and interleukin-2 content was measured with a biotin–avidin system enzyme-linked immunosorbent assay (PharMingen, San Diego, CA, USA). In addition, interleukin-2 protein was also measured in seven samples that were incubated with additional 5 μg/ml anti-interleukin-10 neutralizing antibody (BD PharMingen, San Diego, CA, USA).

2.3. Proliferation assay

Proliferation rates were estimated after 48 h of cultivation. [^{3}H]thymidine (1 μCi, sp. Act. 5 Ci mmol^{-1}, Amersham Pharmacia, Freiburg, Germany) was added for additional 16 h. Cells were harvested onto glass fibre filters (Perkin Elmer, Freiburg, Germany) and the incorporation of [^{3}H] thymidine was determined by liquid scintillation spectroscopy (Perkin Elmer). All cultures were performed in triplicate. Individual results were expressed as background corrected cpm (Δcpm = average cpm of samples − average cpm of media background).

2.4. Flow cytometry

We analyzed the expression of the surface markers of T lymphocytes that were dual stained with the following fluorescent chromogen-conjugated monoclonal antibodies (Becton Dickinson, Heidelberg, Germany): CD3(FITC)/CD25(PE), CD3(FITC)/CD45RO(PE), CD3(PE)/HLA DR(FITC), CD3(PE)/CD54(FITC), CD3(PE)/CD71(FITC) and appropriate isotype-matched control IgG (FITC = fluorescein isothiocyanate; PE = phycoerythrin). Using Interleukin-10 Secretion Assay Kit, Magnetic Cell Sorting (MACS) Separator and MACS Separation Column Type LS (Miltenyi Biotec, Bergisch Gladbach, Germany), we enriched interleukin-10-secreting cells in 1.0×10^8 peri-

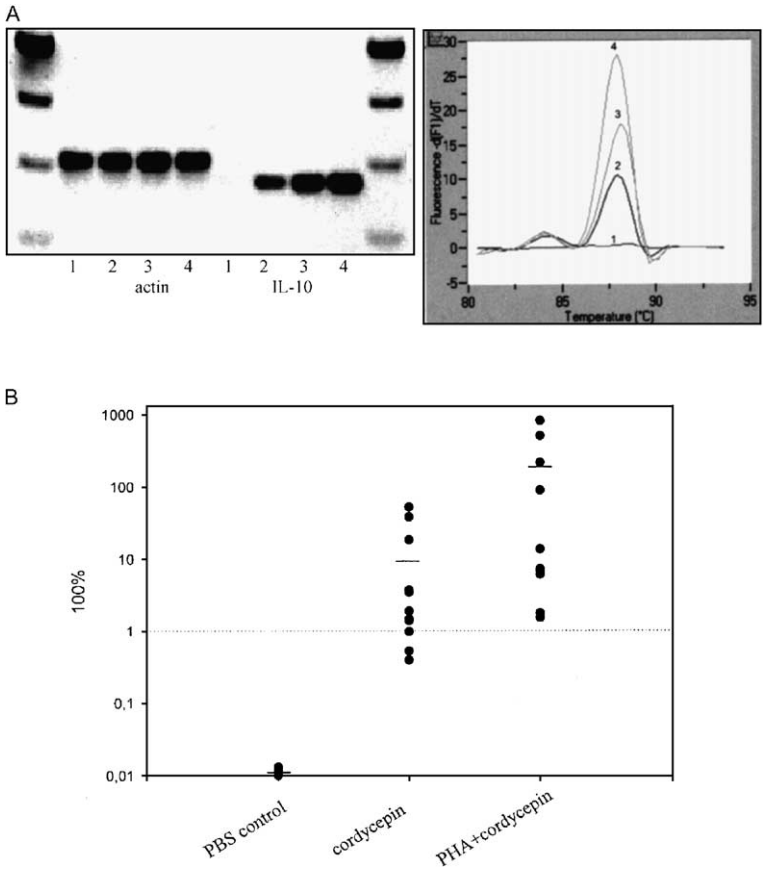

Fig. 2. (A) Comparison of interleukin-10 mRNA products among the samples incubated with PBS (1), cordycepin (2), phytohaemagglutinin (3) and phytohaemagglutinin + cordycepin (4) by the light cycler PCR (right panel) and the agarose gel electrophoresis (left panel). (B) Amounts of interleukin-10 mRNA from untreated control samples and samples treated with cordycepin or phytohaemagglutinin + cordycepin are normalized against the value from the samples treated with phytohaemagglutinin ($n = 10$).

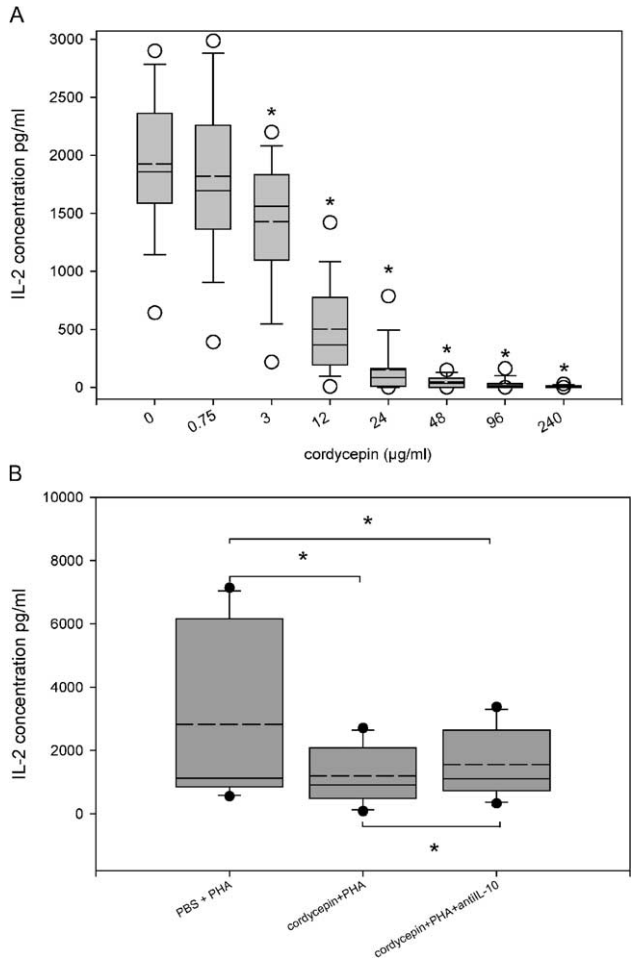

Fig. 3. (A) Phytohaemagglutinin-induced interleukin-2 production significantly declined with the increase of concentration of cordycepin ($n = 11$, $*P < 0.05$). (B) Anti-interleukin-10 neutralizing antibody could not completely block the suppressive effect of cordycepin on phytohaemagglutinin-induced interleukin-2 production ($n = 7$, $*P < 0.05$).

pheral blood mononuclear cells, in order to characterise interleukin-10-secreting lymphocyte subsets. Enriched interleukin-10-secreting cells were separately stained with CD4(FITC), CD8(FITC), CD19(FITC), CD56(FITC) or CD14(FITC).

Apoptosis and death rate of cells were detected with annexin V-FITC and propidium iodide (Annecxin V-FITC Kit, Immunotech, Marseille France).

All samples were analysed by cytofluography on an average of 10,000 events using a flow cytometry FacsCalibur and software Cell Quest (Becton Dickinson).

2.5. Statistical analysis

Data were presented as mean ($\pm$ S.E.M.). The cytokine production and cell proliferation of the samples treated with cordycepin in seven different concentrations were compared to control using a one-way RM analysis (Sigmastat, SPSS, USA). The comparison of parameters from the samples cultured in the presence or absence of cordycepin were done by using a paired *t*-test (Sigmastat).

3. Results

3.1. Interleukin-10 secretion was upregulated by cordycepin

In all samples from 11 subjects, cordycepin showed a strong impact on the production of interleukin-10 by peripheral blood mononuclear cells, even at a low concentration of 0.75 μg/ml, which induced a markedly elevated secretion of interleukin-10. Cordycepin (24 μg/ml) presented a maximal effect with 334.2 ± 75.9 pg/ml of interleukin-10 compared to 0.6 ± 0.5 pg/ml in the absence of cordycepin (Fig. 1A). Only baseline levels of interleukin-10 were detected with 240 μg/ml cordycepin. Even in the presence of phytohaemagglutinin, the nucleoside stimulated a considerably elevated interleukin-10 secretion as compared to appropriate controls (714.8 ± 78.7 vs. $295.6 \pm$

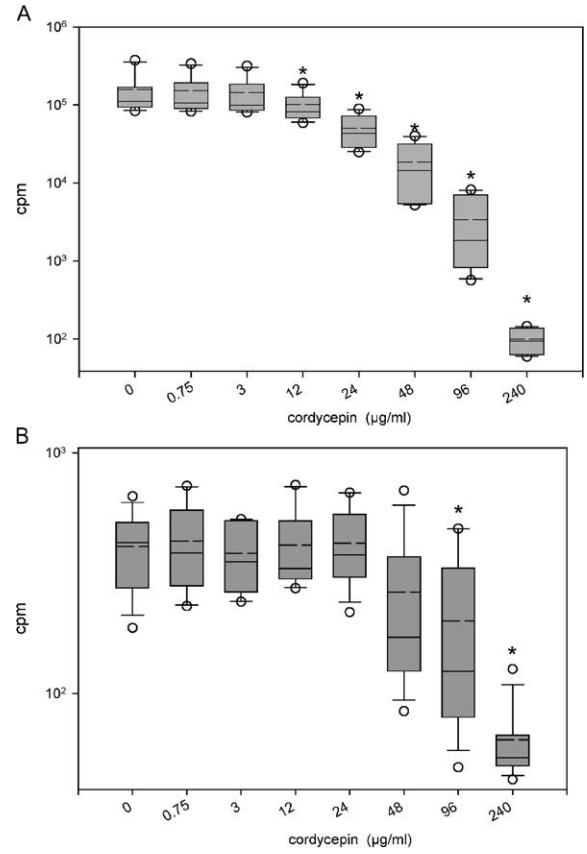

Fig. 4. (A) [^{3}H]thymidine uptake of phytohaemagglutinin-stimulated cells was remarkably reduced by cordycepin and related to cordycepin concentrations ($n = 8$, $*P < 0.05$). (B) Only 96 and 240 μg/ml of cordycepin significantly reduced [^{3}H]thymidine uptake of the cells which were not stimulated by phytohaemagglutinin ($n = 8$, $*P < 0.05$).

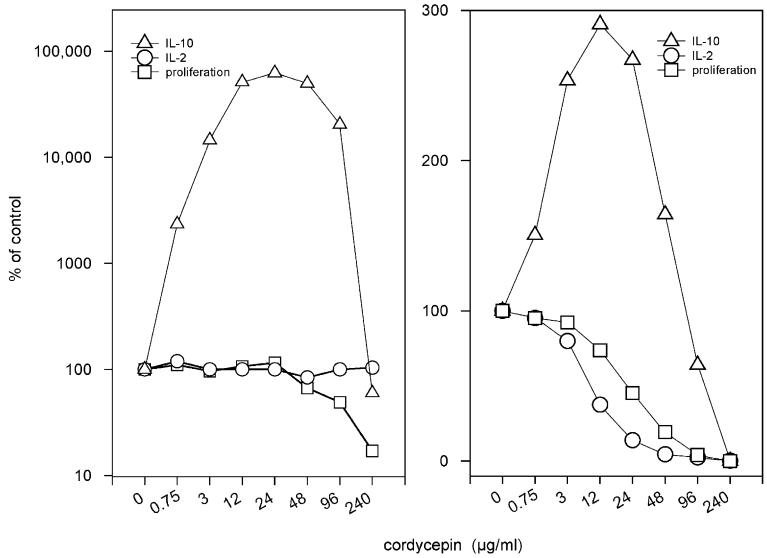

Fig. 5. Relative changes of interleukin-2, interleukin-10 and proliferation with different concentrations of cordycepin. (The percentage of control value was taken as 100%.) Left: without phytohaemagglutinin stimulation. Right: with phytohaemagglutinin stimulation.

57.7 pg/ml). In the presence of phytohaemagglutinin, the maximal effect of cordycepin was seen at 12 μg/ml (Fig. 1B).

3.2. Interleukin-10 mRNA expression was enhanced by cordycepin

Reverse transcriptase-polymerase chain reaction (RT-PCR) was performed to measure the interleukin-10 mRNA expression in samples from 10 subjects. Cordycepin (24 μg/ml) increased the interleukin-10 mRNA expression of all samples, regardless of the presence of phytohaemagglutinin (Fig. 2A). These results showed that without phytohaemagglutinin stimulation, no interleukin-10 mRNA product from the control was detected, but cordycepin was able to induce the expression of interleukin-10 mRNA, and that the calibrated amount of interleukin-10 mRNA in the sample incubated with phytohaemagglutinin and cordycepin was 2–820 times higher than in the sample incubated only with phytohaemagglutinin (Fig. 2B).

3.3. Sorting of interleukin-10-secreting cells

In the samples incubated with cordycepin (24 μg/ml), CD14+ cells account for 44.0 ± 9.0% of the total enriched interleukin-10-secreting cells. The rest were mostly lymphocytes which consist of CD4+ (40.4 ± 3.2%), CD8+ (27.3 ± 4.6%), CD19+ (17.2 ± 6.1%) and CD56+ (58 ± 2.3%) cells.

3.4. Cordycepin inhibits interleukin-2 production

In contrast to the upregulative effect on interleukin-10, cordycepin downregulated the production of interleukin-2. Cordycepin concentrations above 3 μg/ml suppressed phytohaemagglutinin-induced interleukin-2 production markedly compared to control values. The inhibition presents a dose–effect relationship (Fig. 3A). This suppressive effect on interleukin-2 was not completely blocked by anti-interleukin-10 neutralizing antibody; nevertheless, the concentration of interleukin-2 in the samples with anti-interleukin-10 neutralizing antibody was slightly higher than in the PBS control (Fig. 3B). Without phytohaemagglutinin stimulation, the interleukin-2 secretion was very low, regardless of the presence or absence of cordycepin (data not shown).

3.5. Inhibition of proliferative response by cordycepin

Both in the presence and the absence of phytohaemagglutinin, cordycepin reduced the [^{3}H]thymidine uptake of human peripheral blood mononuclear cells. Phytohaemagglutinin-stimulated cells were more susceptible to the inhibitive effect of cordycepin than resting cells. Between 12 and 240 μg/ml, all concentrations of cordycepin significantly inhibited the proliferative response of phytohaemagglutinin-stimulated cells, and the change was similar to that in the production of phytohaemagglutinin-induced interleukin-2 (Fig. 4A). However, in the absence of phytohaemagglutinin, [^{3}H]thymidine uptake of resting cells was influenced only by very high concentrations (96 and 240 μg/ml) of cordycepin (Fig. 4B).

Table 1
Expression of surface markers on T lymphocytes ($n = 9$)

	PBS	Cordycepin	P
CD25	81.6 ± 5.7	74.9 ± 6.5	0.003
CD45RO	64.5 ± 4.6	60.7 ± 4.4	0.05
HLA DR	77.1 ± 3.0	69.3 ± 3.5	0.01
CD54	74.2 ± 5.9	60.5 ± 5.8	0.001
CD71	63.5 ± 5.8	52.9 ± 6.6	0.002

Table 2
Cytotoxicity of cordycepin ($n=11$)

	PBS	Cordycepin	P
Death rate (%)	11.1 ± 1.2	15.0 ± 1.6	0.02
Apoptosis rate (%)	5.0 ± 1.9	6.1 ± 2.2	0.16

The relative changes of the parameters of interleukin-10 production, interleukin-2 production and proliferative response by human peripheral blood mononuclear cells are illustrated in Fig. 5, expressed as a percentage of PBS control. Cordycepin concentration in the range of 0–24 μg/ml showed its upregulative effect on interleukin-10 and downregulative effect on interleukin-2 and cell proliferation.

3.6. Inhibition of expression of surface markers on T lymphocytes by cordycepin

After 24 h of incubation with 24 μg/ml cordycepin or PBS, surface markers CD25, CD45RO, CD54, CD71 and HLA DR antigen expression on phytohaemagglutinin-stimulated T Lymphocytes were determined using flow cytometric analysis. The results showed that 24 μg/ml of cordycepin (the effective concentration that readily influences the production of interleukin-2 or interleukin-10 and the proliferation of peripheral blood mononuclear cells) was able to significantly suppress the expression of these antigens on phytohaemagglutinin-stimulated T lymphocytes (Table 1).

3.7. Cytotoxicity of cordycepin

Effective concentration (24 μg/ml) of cordycepin reduced viability of peripheral blood mononuclear cells in vitro. Death rate of cordycepin 48 h treated cells was 15.0 ± 1.6% vs. 11.1 ± 1.2% in the control ($P=0.02$). However, exposed phosphatidylserine stained by annexin V indicated that cordycepin did not induced more apoptosis than control (Table 2).

4. Discussion

Cordycepin is a nucleoside derivative, 3′-deoxyadenosine, produced from fungi of the genus cordyceps. Various biological effects have been attributed to cordycepin, e.g. inhibition of DNA and RNA synthesis, post transcriptional processing of hn mRNA (Shen, 1991; Pogo, 1974) and activation of adenylate cyclase (Brattin and Portanova, 1978; Stoppini and Baertschi, 1984), enhancement of cell differentiation (Mathew et al., 1989), anti-tumorigenic activity on some cell lines (Deitch and Sawicki, 1979), inhibition of chemotaxis and specific protein synthesis of macrophage cell lines (Plagemann, 1971), anti-fungal activity (Sugar and McCaffrey, 1998) and anti-malarial activity (Trigg et al., 1971). Furthermore, the traditional Chinese herbal medicine "Dong Chong Xia Cao" contains cordycepin, which has been considered as one of the effective components of this herbal drug (Chen and Chu, 1996; Zhu et al., 1998).

Our report documents for the first time that cordycepin shows specific immunomodulating effects. We present evidence that cordycepin could strongly stimulate interleukin-10 production in vitro in a dose-dependent manner and that cordycepin augments the phytohaemagglutinin-induced interleukin-10 production. At the same time, cordycepin inhibited phytohaemagglutinin-induced interleukin-2 production. In our in vitro model, cordycepin has clearly changed the ratio of interleukin-10 vs. interleukin-2. In addition to the influence on cytokine secretion pattern, the effect of cordycepin is reflected by the evidently inhibited proliferative response and T lymphocyte activation. We have attempted to demonstrate whether cordycepin decreased interleukin-2 throughout a raised interleukin-10 level. In our investigation, anti-interleukin-10 neutralizing antibody interfered in the suppressive effect of cordycepin on interleukin-2 production but could not completely block it. This phenomenon indicates that cordycepin reduced the production of interleukin-2 at least in two ways, namely through direct inhibition by cordycepin and indirectly by inducing anti-Th1 cytokine.

Cytokines play a key role in determining the balance between normal host resistance and immunopathology. Polarization of Th1/Th2 balance in the immune system towards the Th1-type of reactivity is involved in susceptibility to autoimmunity, particularly in organ-specific autoimmune diseases (Sher et al., 1998; Eralinna et al., 1998). Evidence from animal models and clinical data suggests that Th1 lymphokines are involved in autoimmune diseases such as experimental uveitis, experimental allergic encephalomylitis and insulin-dependent diabetes mellitus (Singh et al., 1999; Egwuagu et al., 1999; Rapoport et al., 1998; Maron et al., 1999). Recent data indicate that interleukin-10 protects the host against excessive immunopathology (Stohlman et al., 1999; Burkhart et al., 1999; Sun et al., 1999), because it strongly suppresses the functions of Th1 lymphocytes and monocytes and, moreover, decreases the production of inflammatory mediators and proinflammatory cytokines.

Owing to the important modulative functions of interleukin-10 on immune balance, exogenous interleukin-10 has been successfully used to treat some chronic inflammatory diseases, autoimmune diseases and endotoxic shock in animal models (Quinn et al., 2000; Bean et al., 1993; Eppinger et al., 1996; Toto et al., 2000), and in some pilot preclinical investigations, administration of interleukin-10 has shown very promising therapeutic effects on patients with Crohn's disease and chronic hepatitis C. In one study, 46 patients with active steroid-resistant Crohn's disease were treated with recombinant human interleukin-10 intravenously. The treatment was safe, well tolerated and clinically efficacious, and half of the patients experienced a complete remission in the 3-week follow-up period (Van

Deventer et al., 1997). In chronic hepatitis C, short-term therapy with interleukin-10 was well tolerated and caused transient normalization of alanine aminotransferase (ALT, a marker of hepatic inflammation) values in 50% of patients (McHutchison et al., 1999).

Several studies have reported on the attempt to develop agents which could stimulate endogenous interleukin-10 production for the treatment of inflammatory diseases (Ross et al., 1997; Bourrie et al., 1996; Hill et al., 1998). We showed that cordycepin selectively induces the production of interleukin-10 and at the same time inhibits interleukin-2 expression. This mechanism is different from most traditional immunosuppressives which unselectively inhibit immune functions. Therefore, this compound may provide new immunoregulative mechanism and may be beneficial to polytherapeutic schemes for the treatment of autoimmune and inflammatory diseases.

Because of the changes of interleukin-2 and interleukin-10 induced by cordycepin and its other immunosuppressive effects, cordycepin may contribute to a partial explanation of the mechanism of the reported therapeutical effects of "Dong Chong Xia Cao" on some diseases, like chronic nephritis, rheumatoid arthritis, asthma and inflammatory reaction. "Dong Chong Xia Cao" has been used in China for several hundred years. In traditional Chinese medicine, it is believed that "Dong Chong Xia Cao" possesses the ability to invigorate and reinforce the vital essence of kidney and lung.

The cytotoxicity of cordycepin has been investigated mainly in tumor cell lines and terminal deoxynucleotidyl transferase-positive leukemic cells (Koc and McCaffrey, 1995, 1996; Kaufmann, 1991). In our investigation, cordycepin in effective concentration slightly damaged the viability of cells in vitro. However, it has been reported that the administration of cordycepin alone did not induce drug-related toxicities in animal models (Rodman et al., 1997). This is also supported by the observation that "Dong Chong Xia Cao", which contains cordycepin, showed no toxicity to experimental animals and clinical patients (Xu, 1992; Huang et al., 1987).

In this study, a high concentration of cordycepin may have resulted in cytotoxicity or exerted an extensively inhibitory influence on DNA synthesis. However, in the range between 0 and 24 μg/ml of cordycepin, the vigorous upregulation of interleukin-10 expression and simultaneous downregulation of interleukin-2 expression indicate that a cytotoxicity mechanism or an inhibitory effect on DNA synthesis might not be important.

We have demonstrated that the interleukin-10 mRNA content was increased by cordycepin both in phytohaemagglutinin-stimulated cells and resting cells, and that the interleukin-10-producing cells were associated with monocytes and almost all lymphocyte subsets which can secrete interleukin-10, but activating signal transduction pathway is yet unknown. Several new studies (Riese et al., 2000; Platzer et al., 1999) have shown that elevated cAMP levels may increase expression of Th2 cytokines via protein kinase A, cAMP response element-binding protein 1 (CREB-1) or cAMP response element (CRE) pathway. However, cordycepin, a known adenylate cyclase inhibitor (Brattin and Portanova, 1978; Stoppini and Baertschi, 1984), seems unlikely to be involved in this process. Explaining the molecular mechanism of actions caused by cordycepin is our subject for further studies.

In conclusion, cordycepin increased interleukin-10 production and the expression of interleukin-10 mRNA. Moreover, it simultaneously reduced the production of interleukin-2, the proliferation of peripheral blood mononuclear cells and the expression of surface markers indicative for activation of T lymphocytes. In view of the interesting immunoregulative effects of cordycepin, further pharmacological and toxicological investigations are well worth being performed.

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